

Module 2 — Bulk RNA-seq and Gene Expression: Runbook

This is an **operational** document — step-by-step execution instructions, expected outputs, and troubleshooting. For the conceptual "why," see this module's [README.md](#). For the full curriculum context, see [docs/curriculum/modular_bioinformatics_curriculum.md](#).

Prerequisites

- **Tier A (wet-lab pipeline):** access to an HPC cluster with Slurm (sbatch on PATH). Modules available: SRA-Toolkit/3.0.3-gompi-2022a, FastQC/0.11.9-Java-11, fastp/0.23.4-GCC-13.2.0, HISAT2/2.2.1-gompi-2022a, SAMtools/1.16.1-GCC-11.3.0, Subread/2.0.6-GCC-12.3.0, MultiQC/1.28-foss-2024a. A scratch directory with headroom for a downloaded genome + index + BAMs per organism (tens of GB for a mammalian genome; far less for bacterial/viral).
- **Tier B (statistical analysis):** R (≥ 4.0) with DESeq2, ggplot2, pheatmap, RColorBrewer installed, and apeglm if you want the higher-quality shrinkage estimator (script 6 falls back gracefully to normal shrinkage if apeglm is not installed — this is a quality difference, not a hard requirement). Can be run on a laptop with RStudio; no Slurm needed.
- Module 1 completed or reviewed — Tier B assumes the same "validate before you trust it" habit taught there, one level up the pipeline (count matrix instead of raw reads).

Execution order

Run Tier A first to produce a count matrix, then Tier B scripts 1→7 in order from the module's `scripts/` directory.

Step	Command	Where	Expected result
A1	<code>bash smoke_test_prjna1518998.sh</code> (or <code>sbatch</code>)	compute	2-sample smoke test completes; alignment rate reported (target >80%)
A2	<code>sbatch master_rnaseq_pipeline_consolidated.sh</code> (full <code>srr_list.txt</code>)	compute	<code>counts/gene_counts_clean.tsv</code> , <code>final_multiqc_report.html</code>
A3	Write <code>metadata.csv</code> (<code>sample,condition[,batch,...]</code>) matching the count matrix's sample columns	login node	<code>metadata.csv</code> in the working directory

Step	Command	Where	Expected result
B1	Rscript 1_Validate_Counts_And_Metadata.R counts.txt metadata.csv	login/laptop	Console PASS/WARN/FAIL validation report; stops on first hard failure
B2	Rscript 2_Assess_Design_And_Confounding.R metadata.csv	login/laptop	Replicate-count table; condition-vs-column confounding verdict per extra metadata column
B3	Rscript 3_Filter_Low_Count_Genes.R counts.txt metadata.csv	login/laptop	counts_filtered.tsv; before/after gene-count comparison printed
B4	Rscript 4_Exploratory_Analysis_VST_PCA.R counts_filtered.tsv metadata.csv	login/laptop	pca_plot.pdf, sample_distance_heatmap.pdf; any PCA outlier flagged
B5	Rscript 5_Specify_Design_Formula_And_Contrasts.R counts_filtered.tsv metadata.csv <reference_level>	login/laptop	dds_prepared.rds; printed design formula, reference level, and named contrast
B6	Rscript 6_Run_DE_Test_With_Shrinkage_And_FDR.R dds_prepared.rds	login/laptop	deseq2_results.csv, deseq2_results_shrunk.csv; printed FDR-based gene count
B7	Rscript 7_Accessible_Figures_And_Report.R deseq2_results_shrunk.csv	login/laptop	volcano_plot.pdf, ma_plot.pdf, top_genes.csv, DE_REPORT.md
B8	Fill in Section 8 (interpretation) and Section 9 (limitations) of DE_REPORT.md by hand	login/laptop	Completed portfolio artifact

Steps B1-B7 must run in order — each reads the previous step's output file.

Step B5's third argument (reference/control level) matters: omitting it defaults to whichever condition label sorts first alphabetically, which the script itself warns about and which may not be the biological control.

Expected outputs, by step

- **A1-A2:** raw_reads/, trimmed_reads/, alignment/*_sorted.bam, counts/gene_counts_raw.txt, counts/gene_counts_clean.tsv, logs/reference_manifest.txt, final_multiqc_report.html.
- **B1:** stdout only (no new files) — validation report.
- **B2:** stdout only — replicate and confounding report.
- **B3:** counts_filtered.tsv.
- **B4:** pca_plot.pdf, sample_distance_heatmap.pdf.
- **B5:** dds_prepared.rds.
- **B6:** deseq2_results.csv, deseq2_results_shrunk.csv.
- **B7:** volcano_plot.pdf, ma_plot.pdf, top_genes.csv, DE_REPORT.md.

Troubleshooting

Symptom	Likely cause	Fix
Script B1 fails: "count matrix contains non-numeric columns"	Input is not a raw featureCounts/count file (extra text columns, wrong delimiter)	Confirm the file is tab-separated with a gene-ID first column; re-export from Tier A if unsure
Script B1 fails: "N values are non-integer... looks like normalized data"	You passed TPM/FPKM/CPM instead of raw counts	Re-run from <code>gene_counts_clean.tsv</code> (raw featureCounts output), not a normalized export
Script B1 fails: "count columns with no metadata row" / "metadata rows with no matching count column"	Sample names in <code>metadata.csv</code> don't exactly match the count matrix's column headers (case, extra suffix, etc.)	Compare <code>colnames(counts)</code> to <code>metadata.csv</code> 's <code>sample</code> column character-for-character
Script B2 reports [FAIL] ' <code><column></code> ' is PERFECTLY CONFOUNDED with condition	Every sample in one condition group was also in only one batch/sex/etc. level	This is a design problem, not a statistics problem — the only real fix is re-sequencing with that variable balanced across conditions; do not attempt to "fix" it by adding the column to the design formula anyway
Script B3's documented rule keeps fewer genes than the naive rule	Working as intended — genes concentrated in a single sample (not spread across a group) are being correctly excluded; inspect the specific gene IDs flagged in the console output if you want to confirm	
Script B4 errors on <code>vst()</code> with "less than 'nsub' rows"	Fewer than 1000 genes remain after filtering (small gene panel, bacterial/viral genome, or aggressive filtering)	Already handled — the script caps <code>nsub</code> at the actual gene count automatically; if you still see this error, you are running an older/unpatched copy of the script
Script B4's PCA legend shows literal R code (e.g. <code>if (has_batch) batch else NULL</code>) instead of "batch"	Running an older/unpatched copy of the script	Use the current version, which builds the shape aesthetic conditionally rather than inline inside <code>aes()</code>
Script B5 errors: "requested reference level '...' is not among the condition levels found"	Typo in the reference-level argument, or it doesn't match <code>metadata.csv</code> 's condition labels exactly	Check the exact condition labels printed by script B2 and pass one of those verbatim
Script B6 prints [WARN] <code>apeglm</code> shrinkage unavailable	<code>apeglm</code> package is not installed	Expected fallback to <code>normal</code> shrinkage — install <code>apeglm</code> via <code>BiocManager::install("apeglm")</code> for the preferred estimator, or accept the fallback for teaching purposes
Script B6 reports 0 significant genes when you expected some	Underpowered design (see script B2's replicate-count warning), a genuinely null result, or the wrong reference level (check the printed contrast direction)	Re-check script B2's replication warning and script B5's printed contrast before assuming the pipeline is broken

Symptom	Likely cause	Fix
Script B7's DE_REPORT.md Sections 8-9 are blank	Expected — these are intentionally left for the analyst to complete by hand	Fill them in based on the actual top-gene biology and the caveats surfaced by scripts B2, B4, and B6

Completion checklist (portfolio artifact)

The curriculum's Module 2 assessment asks for: *"a reproducible report containing metadata validation, exploratory plots, model specification, differential-expression results, effect-size visualization, interpretation, and limitations."* Confirm each is present in DE_REPORT.md before considering the module complete:

- ☐ **Metadata validation** — Section 1 reflects a real script B1 PASS report (or documents and resolves any WARN), not a placeholder.
- ☐ **Design and confounding check** — Section 2 states the actual replicate count per condition and the confounding verdict for every extra metadata column from script B2 — with any [FAIL] resolved before proceeding.
- ☐ **Gene filtering rule** — Section 3 states the exact documented rule and the real before/after gene counts from script B3, not a generic description.
- ☐ **Exploratory plots** — Section 4 references real, opened pca_plot.pdf and sample_distance_heatmap.pdf files, with any flagged PCA outlier addressed (excluded with a stated reason, or kept with a stated justification).
- ☐ **Model specification** — Section 5 states the exact design formula, reference level, and named contrast from script B5 — never "the default."
- ☐ **Differential-expression results** — Section 6 reports the FDR-based (padj) significant-gene count, never a raw-p-value count.
- ☐ **Effect-size visualization** — Section 7 references the real volcano_plot.pdf/ma_plot.pdf, both using shrunk log2FoldChange values.
- ☐ **Interpretation** — Section 8 is filled in by hand with a real biological reading of the top hits, not left as the auto-generated prompt text.
- ☐ **Limitations** — Section 9 is filled in by hand, explicitly stating the actual replicate count, any excluded outliers, and any significant-but-trivial-effect genes excluded from the narrative.